

Protocol Sequences for Age of Information under Multiple-Packet Reception

Yinian Zheng*, Fang Liu[†], Yuan-Hsun Lo[‡], Tsai-Lien Wong[§], Kangkang Xu[§] and Yijin Zhang*

*School of Electronic and Optical Engineering, Nanjing University of Science and Technology, Nanjing, China

[†]Department of Information Engineering, The Chinese University of Hong Kong, Hong Kong, China

[‡]Department of Applied Mathematics, National Pingtung University, Pingtung, Taiwan

[§]Department of Applied Mathematics, National Sun Yat-sen University, Kaohsiung, Taiwan

Email: yinian.zheng@njust.edu.cn, fangliu@cuhk.edu.hk, yhlo0830@gmail.com,

tlwong@math.nsysu.edu.tw, ivykkxu107@gmail.com, yijin.zhang@gmail.com.

Abstract—This paper focuses on protocol sequences for age of information (AoI) in a multiple-packet reception (MPR) channel without feedback and synchronization. Unlike traditional probabilistic schemes, protocol-sequences-based schemes allow each user to deterministically decide when to transmit only according to its assigned sequence. When the MPR capability $\gamma = 2$, we use a previously known construction to generate user-irrepressible (UI) sequences that are favorable for the AoI improvement. Under this construction, by studying the reverse Hamming cross-correlations of the corresponding sequences, which is more complicated than that in prior studies, we provide an analytical approach for evaluating the AoI for $\gamma = 2$. When $\gamma > 2$, we also propose a new construction to produce UI sequences for the AoI improvement. Simulation results show that the proposed schemes outperform slotted ALOHA in terms of average AoI and worst-case AoI for various settings.

I. INTRODUCTION

Information freshness optimization is essential to support real-time communication, sensing and control for mission-critical applications in Internet of Things (IoT). Fundamentally different from the conventional latency metric that measures the end-to-end delivery time of a packet, age of information (AoI), defined as the elapsed time since the moment that the freshest delivered update was generated, has been introduced in [1] to measure the information freshness. In this paper, we focus on the AoI under multiple access without feedback and synchronization, which is a common scenario in low-cost and low-energy monitoring IoT systems.

The AoI under probabilistic multiple access has gained much research attention in recent years. In the absence of feedback, [2] investigated the AoI under slotted ALOHA where each user adopts a fixed transmission probability. Under immediate feedback and full synchronization, [3] analyzed the AoI under dynamic slotted ALOHA where each user adjusts its transmission probability according to the contention level, but assumed an unrealistic channel observation. Further, still under immediate feedback, [4] designed an age-dependent slotted ALOHA scheme but required each user to know whether its instantaneous age exceeds a threshold.

Unlike traditional probabilistic schemes, in the absence of synchronization and feedback, a protocol-sequences-based scheme, where each user is assigned a periodic binary sequence from the set of CRT sequences [5] to determine when

to transmit, was proposed in [6]. Owing to the appropriately designed sequence structure especially with the user-irrepressible (UI) property [7], which ensures at least one collision-free transmission from each user within a sequence period, such a scheme achieves better AoI performance than optimal slotted ALOHA. It was also shown in [6] that the analytical modeling for the AoI evaluation under a protocol-sequences-based scheme is quite different from that for the throughput and latency evaluation [8]–[10].

The results reported in [6] inspire a new direction of access scheme design for AoI. However, the single-packet reception (SPR) channel (i.e., the traditional collision channel) model adopted in [6] has become somewhat restrictive, due to the advance of multiple-packet reception (MPR) techniques [11] for accommodating multiple concurrent successful transmissions. As the MPR has a significant impact on the construction of UI sequences [12] and the resultant AoI, this paper makes a first attempt to design protocol-sequences-based schemes that are favorable for the AoI improvement under MPR. We present how to construct protocol sequences for different MPR settings and prove the UI property. Further, by studying the reverse Hamming cross-correlations of the corresponding protocol sequences in more detail, which is more complicated than that in [6], we provide an analytical approach for evaluating the AoI. Simulation results show that the proposed schemes outperform optimal slotted ALOHA in terms of average AoI and worst-case AoI in various settings.

Section II describes the system model and definitions related to protocol sequences. We focus on a special MPR channel in Section III, and then focus on more MPR channels in Section IV. Section V presents simulation results and Section VI concludes the paper.

II. SYSTEM MODE AND DEFINITIONS

A. System Description

Consider a fully-connected network where M users want to transmit their status updates to an access point (AP). The channel time is divided into equal time slots. Each user can take a new update at the beginning of each slot and the transmission time of each update occupies exactly a slot.

There is no accurate clock synchronization for low-cost users. Define the time offset of a user as the time difference between the clocks of this user and the AP. For user $i \in \mathbb{Z}_M \triangleq \{0, 1, \dots, M-1\}$, denote its time offset by τ_i , which is an integer multiple of one slot duration. The time offsets of the users remain constant during the transmission session, but are always unknown to each user. From now on, all referred time indexes are understood to be in reference to the AP clock. We will focus on such a slot-synchronous model since the results obtained in this model can be easily extended to the fully asynchronous model [8].

Consider an MPR channel and refer to γ ($1 \leq \gamma < M$) as the MPR capability [11]. Specifically, assume that an update can be received successfully if and only if at most $(\gamma-1)$ other updates are being transmitted simultaneously. Further, assume that the AP does not send any feedback information so that each user keeps transmitting according to its access schedule without knowing the results of its previous transmissions.

Each user i is assigned a unique zero-one protocol sequence $\mathbf{s}_i \triangleq [s_i(0) s_i(1) s_i(2) \dots s_i(L-1)]$, where L is the sequence period, from the set of protocol sequences $\mathcal{S} \triangleq \{\mathbf{s}_0, \mathbf{s}_1, \dots, \mathbf{s}_{M-1}\}$. Then, at the beginning of each slot t , each user i transmits an update if and only if $s_i(t \oplus \tau_i) = 1$ where $\oplus$ denotes the modulo- L addition. Note that the design of $\mathcal{S}$ depends on both the number of users M and the MPR capability γ , and will be introduced in subsequent sections.

B. Evolution of the AoI

Let $A_i(t)$ denote the instantaneous AoI of user i at the beginning of slot t . Then, due to the considered generate-at-will traffic, we have

$$A_i(t) \triangleq \begin{cases} 1, & \text{if user } i \text{ transmits successfully} \\ & \text{at slot } t, \\ A_i(t-1) + 1, & \text{otherwise.} \end{cases} \quad (1)$$

The average AoI of user i , denoted by $\overline{A}_i$, is defined as

$$\overline{A}_i \triangleq \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^T A_i(t). \quad (2)$$

C. More Definitions

We introduce more definitions for protocol sequences that will be useful. Define the *characteristic set* of $\mathbf{s}_i$ as

$$\mathcal{I}_{\mathbf{s}_i} \triangleq \{t \in \mathbb{Z}_L | s_i(t) = 1\}$$

where $\mathbb{Z}_L \triangleq \{0, 1, \dots, L-1\}$. The *Hamming weight* of $\mathbf{s}_i$ is $|\mathcal{I}_{\mathbf{s}_i}|$. Define $\mathcal{I}_{\mathbf{s}_i}^{\tau_i} \triangleq \mathcal{I}_{\mathbf{s}_i} \oplus \tau_i$. For two distinct sequences $\mathbf{s}_i$ and $\mathbf{s}_j$, their *Hamming cross-correlation* with the relative time offset τ is defined by

$$H_{\mathbf{s}_i, \mathbf{s}_j}(\tau) \triangleq |\mathcal{I}_{\mathbf{s}_i} \cap \mathcal{I}_{\mathbf{s}_j}^{\tau}|. \quad (3)$$

Further, define $H_{\mathbf{s}_i, \mathbf{s}_j}$ as the maximum value of $H_{\mathbf{s}_i, \mathbf{s}_j}(\tau)$ over all possible $\tau \in \mathbb{Z}_L$.

A set of sequences $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_{M-1}\}$ is called UI if for any $i \in \mathbb{Z}_M$, at least one “1” in $\mathbf{s}_i$ is blocked by at most $\gamma-1$ “1”s in other sequences for all possible time offsets.

III. THE CASE OF MPR CAPABILITY $\gamma = 2$

This section first describes the used sequence construction for the MPR capability $\gamma = 2$. Based on this construction, we investigate the time offsets corresponding to different collision positions, and use these information to derive the *cumulative distribution function* (CDF) of AoI and average AoI.

A. Sequence Construction

For the MPR capability $\gamma = 2$, we use *prime sequences* [13] as protocol sequences. The construction method is given as follows.

Construction 1 ([13]). *Let p be a prime not smaller than M . For $i \in \mathbb{Z}_M$, the characteristic set of $\mathbf{s}_i$ is*

$$\mathcal{I}_{\mathbf{s}_i} = \{i \cdot k \pmod{p} + kp : k = 0, 1, \dots, p-1\}. \quad (4)$$

For example, when $p = 5$, the five characteristic sets are

$$\begin{aligned} \mathcal{I}_{\mathbf{s}_0} &= \{0, 5, 10, 15, 20\}, \quad \mathcal{I}_{\mathbf{s}_1} = \{0, 6, 12, 18, 24\}, \\ \mathcal{I}_{\mathbf{s}_2} &= \{0, 7, 14, 16, 23\}, \quad \mathcal{I}_{\mathbf{s}_3} = \{0, 8, 11, 19, 22\}, \\ \mathcal{I}_{\mathbf{s}_4} &= \{0, 9, 13, 17, 21\}. \end{aligned}$$

Construction 1 can produce p sequences, each of which is of period p^2 and Hamming weight p . It was proved in [13] that $H_{\mathbf{s}_i, \mathbf{s}_j} = 2$ for distinct $i, j \in \mathbb{Z}_M \setminus \{0\}$ and $H_{\mathbf{s}_0, \mathbf{s}_j} = 1$ for $j \in \mathbb{Z}_M \setminus \{0\}$. We further obtain the following UI property.

Lemma 1. *The set of sequences produced by Construction 1 is UI for the MPR capability $\gamma = 2$.*

Proof. For each sequence, the number of overlappings is at most $2(M-1)$. Since the Hamming weight is $p \geq M$, by the Pigeonhole Principle, there exists at least one transmission which is blocked by other sequences at most one time. Hence, the UI property holds under the MPR capability $\gamma = 2$. $\square$

Such a UI property implies that the worst-case AoI under $\gamma = 2$ must be equal to or smaller than the sequence period $L = p^2$.

B. Time Offsets Classification

For a characteristic set $\mathcal{I} \subseteq \mathbb{Z}_L$, define

$$\Delta(\mathcal{I}) \triangleq \{x \ominus y : x, y \in \mathcal{I}, x \neq y\}$$

where $\ominus$ denotes the modulo- L subtraction. It is clear that $H_{\mathbf{s}_i, \mathbf{s}_j} = 1$ if and only if $\Delta(\mathcal{I}_{\mathbf{s}_i}) \cap \Delta(\mathcal{I}_{\mathbf{s}_j}) = \emptyset$, and $H_{\mathbf{s}_i, \mathbf{s}_j} = 2$ if and only if $\Delta(\mathcal{I}_{\mathbf{s}_i}) \cap \Delta(\mathcal{I}_{\mathbf{s}_j}) = \{\pm d\}$ for some $d \in \mathbb{Z}_L$.

It was shown in [10] that

$$|\{\tau \in \mathbb{Z}_L : H_{\mathbf{s}_i, \mathbf{s}_j}(\tau) > 0\}| = |\mathcal{I}_{\mathbf{s}_i}| |\mathcal{I}_{\mathbf{s}_j}| - e(\mathcal{I}_{\mathbf{s}_i}, \mathcal{I}_{\mathbf{s}_j}), \quad (5)$$

where

$$\begin{aligned} e(\mathcal{I}_{\mathbf{s}_i}, \mathcal{I}_{\mathbf{s}_j}) & \\ & \triangleq \begin{cases} 0, & \text{if } \Delta(\mathcal{I}_{\mathbf{s}_i}) \cap \Delta(\mathcal{I}_{\mathbf{s}_j}) = \emptyset, \\ f_i(\mathcal{I}_{\mathbf{s}_i}, \mathcal{I}_{\mathbf{s}_j}) f_j(\mathcal{I}_{\mathbf{s}_i}, \mathcal{I}_{\mathbf{s}_j}), & \text{if } \Delta(\mathcal{I}_{\mathbf{s}_i}) \cap \Delta(\mathcal{I}_{\mathbf{s}_j}) = \{\pm d\}, \end{cases} \end{aligned}$$

and

$$f_i(\mathcal{I}_{s_i}, \mathcal{I}_{s_j}) \triangleq |\{(x, y) \in \mathcal{I}_{s_i} \times \mathcal{I}_{s_j} : x \ominus y = d\}|,$$

$$f_j(\mathcal{I}_{s_i}, \mathcal{I}_{s_j}) \triangleq |\{(x, y) \in \mathcal{I}_{s_j} \times \mathcal{I}_{s_i} : x \ominus y = d\}|.$$

Assume the time offset of user i is fixed at τ_i^* . For an arbitrary subset $\mathcal{A} \subseteq \mathcal{I}_{s_i}^*$ and an arbitrary $j \in \mathbb{Z}_M \setminus \{i\}$, we define $\alpha_j(\mathcal{A})$, the *reverse Hamming cross-correlation*, as the number of time offsets τ_j so that $\mathcal{I}_{s_i}^{\tau_i^*} \cap \mathcal{I}_{s_j}^{\tau_j} = \mathcal{A}$, i.e.,

$$\alpha_j(\mathcal{A}) \triangleq \left| \left\{ \tau_j \in \mathbb{Z}_L : \mathcal{I}_{s_i}^{\tau_i^*} \cap \mathcal{I}_{s_j}^{\tau_j} = \mathcal{A} \right\} \right|.$$

As $H_{s_i s_j} \leq 2$, it suffices to consider that $|\mathcal{A}| = 0, 1$ or 2 . Note that [6] only needs to consider $|\mathcal{A}| = 0$ or 1 .

The case of $|\mathcal{A}| = 0$. In this case, $\alpha_j(\mathcal{A})$ counts the number of time offsets τ_j such that $\mathcal{I}_{s_i}^{\tau_i^*} \cap \mathcal{I}_{s_j}^{\tau_j} = \emptyset$. By (5), we have

$$\begin{aligned} \alpha_j(\mathcal{A}) &= L - (|\mathcal{I}_{s_i}^{\tau_i^*}| |\mathcal{I}_{s_j}^{\tau_j}| - e(\mathcal{I}_{s_i}^{\tau_i^*}, \mathcal{I}_{s_j}^{\tau_j})) \\ &= L - p^2 + e(\mathcal{I}_{s_i}, \mathcal{I}_{s_j}). \end{aligned}$$

The case of $|\mathcal{A}| = 2$. In this case, assume $\mathcal{A} = \{x_a, x_b\}$ for some $x_a < x_b$. Consider $\Delta(\mathcal{A}) = \{\pm(x_b - x_a)\}$. According to the definition of $\alpha_j(\mathcal{A})$, we have

$$\begin{aligned} \alpha_j(\mathcal{A}) &= \left| \left\{ \tau_j \in \mathbb{Z}_L : \{x_a, x_b\} \subseteq \mathcal{I}_{s_j}^{\tau_j} \right\} \right| \\ &= \left| \left\{ (b_1, b_2) \in \mathcal{I}_{s_j} \times \mathcal{I}_{s_j} : b_1 \ominus b_2 = x_b - x_a \right\} \right| \\ &= f_j(\mathcal{A}, \mathcal{I}_{s_j}). \end{aligned}$$

which indeed counts the number of pairs $(b_1, b_2) \in \mathcal{I}_{s_j} \times \mathcal{I}_{s_j}$ with $b_1 \ominus b_2 = x_b - x_a$.

The case of $|\mathcal{A}| = 1$. In this case, $\alpha_j(\mathcal{A})$ counts the number of time offsets τ_j so that $\mathcal{I}_{s_i}^{\tau_i^*} \cap \mathcal{I}_{s_j}^{\tau_j} = \mathcal{A} = \{x\}$ for some specific $x \in \mathcal{I}_{s_i}^*$. Since each element in $\mathcal{I}_{s_j}$ can meet x_a exactly once under some time offset, it suffices to subtract the number of shifts τ_j such that $|\mathcal{I}_{s_i}^{\tau_i^*} \cap \mathcal{I}_{s_j}^{\tau_j}| = 2$. Hence,

$$\alpha_j(\mathcal{A}) = p - \sum_{x \in \mathcal{I}_{s_i}^* \setminus \mathcal{A}} f_j(\mathcal{A} \cup \{x\}, \mathcal{I}_{s_j}).$$

The previous results can be summarized as follows.

Lemma 2. Consider an arbitrary time offset $\tau_i^* \in \mathbb{Z}_L$ of user i for an arbitrary $i \in \mathbb{Z}_M$. For each $j \in \mathbb{Z}_M \setminus \{i\}$ and each $\mathcal{A} \subseteq \mathcal{I}_{s_i}^*$, we have

$$\alpha_j(\mathcal{A}) = \begin{cases} L - p^2 + e(\mathcal{I}_{s_i}, \mathcal{I}_{s_j}), & \text{if } |\mathcal{A}| = 0, \\ p - \sum_{x \in \mathcal{I}_{s_i}^* \setminus \mathcal{A}} f_j(\mathcal{A} \cup \{x\}, \mathcal{I}_{s_j}), & \text{if } |\mathcal{A}| = 1, \\ f_j(\mathcal{A}, \mathcal{I}_{s_j}), & \text{if } |\mathcal{A}| = 2. \end{cases}$$

C. CDF of AoI

Still fix the time offset of s_i at τ_i^* . Consider an event that the set of collision positions in $\mathcal{I}_{s_i}^{\tau_i^*}$ caused by user j is $\mathcal{B}_j$ for each $j \in \mathbb{Z}_M \setminus \{i\}$. Under an arbitrary chosen combination of time offsets of other $M - 1$ sequences, this event occurs with the probability

$$\prod_{j \in \mathbb{Z}_M \setminus \{i\}} \alpha_j(\mathcal{B}_j) / L^{M-1}. \quad (6)$$

Define the multiset $\mathcal{C} \triangleq \cup_{j \in \mathbb{Z}_M \setminus \{i\}} \mathcal{B}_j$.

Denote by $\mathcal{R}$ the set of successful positions in $\mathcal{I}_{s_i}^*$ under $\mathcal{C}$. For a given set S , let S^2 denote the multiset $S \cup S$. Under the MPR capability $\gamma = 2$, we have

$$\mathcal{R} = \mathcal{I}_{s_i}^* \cap (\mathcal{I}_{s_i}^{*2} \setminus \mathcal{C}). \quad (7)$$

Due to the UI property, for some $m \geq 1$, we can write $\mathcal{R} = \{r_0, r_1, \dots, r_{m-1}\}$ with $0 \leq r_0 < r_1 < \dots < r_{m-1} \leq L - 1$. Furthermore, define the inter-departure vector of $\mathcal{R}$ by $\mathbf{v} \triangleq (v_0 = r_1 - r_0, v_1 = r_2 - r_1, \dots, v_{m-1} = L + r_0 - r_{m-1})$.

Note that each successful transmission will cause $A_i(t)$ to restart at 0 and increase linearly until the next successful transmission. So, if $A_i(t) = \omega$, the inter-departure time until the beginning of slot t must exceed ω . Then, the conditional probability $\Pr(A_i(t) = \omega | \tau_i^*, \mathcal{R})$ can be obtained by calculating the ratio of the entries in $\mathbf{v}$ that exceed ω . Based on this fact, we obtain

$$\begin{aligned} \Pr(A_i(t) \leq \Omega | \tau_i^*, \mathcal{R}) &= \sum_{\omega=0}^{\Omega} \Pr(A_i(t) = \omega | \tau_i^*, \mathcal{R}) \\ &= \sum_{\omega=0}^{\Omega} |\{l \in \mathbb{Z}_m | v_l > \omega\}| / L. \end{aligned} \quad (8)$$

By (7) and (8), we have

$$\begin{aligned} \Pr(A_i(t) \leq \Omega | \tau_i^*) &= \sum_{\mathcal{I}_{s_i}^{*2} \setminus \mathcal{C} \neq \emptyset} \Pr(A_i(t) \leq \Omega | \tau_i^*, \mathcal{R}) \prod_{j \in \mathbb{Z}_M \setminus \{i\}} \alpha_j(\mathcal{B}_j) / L^{M-1}. \end{aligned} \quad (9)$$

Further, by summing (9) over all possible τ_i^* , we obtain the following CDF of AoI.

$$\Pr(A_i(t) \leq \Omega) = \sum_{\tau_i^*=0}^{L-1} \frac{1}{L} \Pr(A_i(t) \leq \Omega | \tau_i^*). \quad (10)$$

D. Average AoI

A straightforward method to derive the average AoI is based on the CDF of AoI obtained in Section III-C. Nevertheless, here, we use the idea in [2] to derive the average AoI directly based on the inter-departure time for reducing the computational complexity.

Let t_i^q denote the slot index when user i successfully transmits its update for the q -th time. Let Y_i^q denote the inter-departure time between the q -th and $(q + 1)$ -th successfully transmitted updates of user i , i.e., $Y_i^q = t_i^{q+1} - t_i^q$. By the definition of Y_i^q , we have

$$\mathbb{E}[Y_i^q | \tau_i^*, \mathcal{R}] = \frac{L}{\left| \mathcal{I}_{s_i}^* \cap (\mathcal{I}_{s_i}^{*2} \setminus \mathcal{C}) \right|}. \quad (11)$$

Given $\mathcal{I} = \{x_0, x_1, \dots, x_{m-1}\}$, define

$$\eta(\mathcal{I}) \triangleq (x_1 - x_0)^2 + (x_2 - x_1)^2 + \dots + (L + x_0 - x_{m-1})^2.$$

Then, based on (7) and the definition of $\eta(\mathcal{I})$, we have

$$\mathbb{E}[(Y_i^q)^2 | \tau_i^*, \mathcal{R}] = \frac{\eta(\mathcal{I}_{s_i}^{\tau_i^*} \cap (\mathcal{I}_{s_i}^{\tau_i^* 2} \setminus \mathcal{C}))}{|\mathcal{I}_{s_i}^{\tau_i^*} \cap (\mathcal{I}_{s_i}^{\tau_i^* 2} \setminus \mathcal{C})|}. \quad (12)$$

According to [2], we can obtain the conditional expected value of $A_i(t)$ by (11) and (12) as

$$\mathbb{E}[A_i(t) | \tau_i^*, \mathcal{R}] = \frac{\mathbb{E}[(Y_i^q)^2 | \tau_i^*, \mathcal{R}]}{2\mathbb{E}[Y_i^q | \tau_i^*, \mathcal{R}]} - \frac{1}{2}. \quad (13)$$

Finally, by summing (13) over all possible τ_i^* and $\mathcal{R}$ with Lemma 2 and (6), we obtain

$$\overline{A}_i = \sum_{\tau_i^*=0}^{L-1} \frac{1}{L} \sum_{\substack{\mathcal{I}_{s_i}^{\tau_i^*} \setminus \mathcal{C} \neq \emptyset}} \mathbb{E}[A_i(t) | \tau_i^*, \mathcal{R}] \prod_{j \in \mathbb{Z}_M \setminus \{i\}} \alpha_j(\mathcal{B}_j) / L^{M-1}. \quad (14)$$

IV. THE CASES OF MPR CAPABILITY $\gamma > 2$

It is difficult to extend the prime sequences to produce favorable access patterns for AoI when the MPR capability $\gamma > 2$. As such, in this section, we provide a suitable sequence construction for an arbitrary $\gamma > 2$, and show that the produced sequence set is UI under the considered γ . However, the AoI performance is not theoretically analyzed here due to the page limit, and will be examined through simulations in Section V.

Based on the previously known *CRT-UI sequences* [14], generalization of *extended prime sequences* [13], we use an iterative algorithm to construct new protocol sequences, called γ -*CRT-UI sequences*. The construction is given as follows.

Construction 2. Let γ be an integer larger than 2, and p be the smallest prime not less than M . Choose an integer $q \geq 2p - 1$ with $\gcd(p, q) = 1$ so that the following holds: there exist $\tau_{i_0}, \tau_{i_1}, \dots, \tau_{i_{\gamma-1}} \in \mathbb{Z}_{pq}$ such that $\mathcal{I}_{g_i}^{\tau_{i_0}}, \mathcal{I}_{g_i}^{\tau_{i_1}}, \dots, \mathcal{I}_{g_i}^{\tau_{i_{\gamma-1}}}$ are mutually disjoint, where

$$\mathcal{I}_{g_i} = \{i \cdot k \pmod{p} + kq : k = 0, 1, \dots, p-1\}, \quad (15)$$

for $i \in \mathbb{Z}_M$. Then, define the characteristic set of s_i as

$$\mathcal{I}_{s_i} = \bigcup_{n=0}^{\gamma-1} \mathcal{I}_{g_i}^{\tau_{i_n}}. \quad (16)$$

For example, when $\gamma = 3$ and $M = 5$, we let $p = 5$ and choose $q = 9$. Let us focus on the construction of s_4 . By (15),

$$\mathcal{I}_{g_4} = \{0, 13, 21, 29, 37\}.$$

And, $\tau_{4_0}, \tau_{4_1}, \tau_{4_2}$ can be chosen as 0, 3 and 5, respectively, so that $\mathcal{I}_{g_4}^0 = \mathcal{I}_{g_4}$ and

$$\begin{aligned} \mathcal{I}_{g_4}^3 &= \{3, 16, 24, 32, 40\}. \\ \mathcal{I}_{g_4}^5 &= \{5, 18, 26, 34, 42\}. \end{aligned}$$

Finally, we get

$$\mathcal{I}_{s_4} = \{0, 3, 5, 13, 16, 18, 21, 24, 26, 29, 32, 34, 37, 40, 42\}.$$

Construction 2 can produce M sequences, each of which is of period pq and Hamming weight γp . It is shown in [14] that $H_{g_i, g_j} = 1$ for distinct $i, j \in \mathbb{Z}_M$. As $\mathcal{I}_{s_i}$ is a disjoint union of γ shifted versions of $\mathcal{I}_{g_i}$, we have $H_{s_i, s_j} \leq \gamma^2$. We further obtain the following UI property.

Lemma 3. Let γ be an arbitrary integer larger than 2. The set of sequences produced by Construction 2 is UI for the MPR capability γ .

Proof. For each sequence, the number of transmission overlappings is at most $\gamma^2(M-1)$, which is strictly less than $\gamma^2 p$. Since the Hamming weight is γp , by the Pigeonhole Principle, there exists at least one transmission which is unblocked or blocked by other sequences at most $\gamma - 1$ times. Hence, the UI property holds under the MPR capability γ . $\square$

Such a UI property implies that the worst-case AoI under the given $\gamma > 2$ must be equal to or smaller than the sequence period $L = pq$. Further, such a UI property together with the sequence structure of Construction 2 are helpful to reducing the average AoI, which will be shown in Section V.

V. SIMULATION RESULTS

In this section, we conduct simulation experiments to evaluate the average AoI and worst-case AoI of the proposed schemes under the MPR capability γ . We consider optimal slotted ALOHA for comparison. We comply with the system model specified in Section II to set up the experiments, and each simulation result is obtained from 10^6 independent experiments under randomly generated time offsets.

A. Comparison under $\gamma = 2$

For $\gamma = 2$ and $M = 7$, Fig. 1 compares the CDF of AoI and average AoI of the proposed scheme and optimal slotted ALOHA. As observed, under the proposed scheme, the analytical and simulative results match well for all the users, which validates the theoretical study in Section II. Moreover, we find that all the users have almost the same CDF and the same average AoI, which indicates good fairness in the proposed scheme. Note that the fairness problem is a major concern in deterministic access. We also see a notable gap between the proposed scheme and optimal slotted ALOHA. Such advantages can be attributed to both the short period and UI property of prime sequences used in the proposed scheme.

For $\gamma = 2$ and different M , Fig. 2 further compares the simulative average AoI of the proposed scheme and optimal slotted ALOHA. We observe that the proposed scheme significantly outperforms optimal slotted ALOHA in all the cases especially when M is close to p in Construction 1. This is because a shorter sequence period leads to more frequently successful updates and is beneficial to reducing the AoI.

For $\gamma = 2$ and different M , Table I shows that the proposed scheme produces much smaller worst-case AoI than optimal slotted ALOHA in all the cases. This advantage verifies the effect of the UI property on the worst-case AoI. This advantage together with the advantage of average AoI indicate that the proposed scheme is suitable for mission-critical applications.

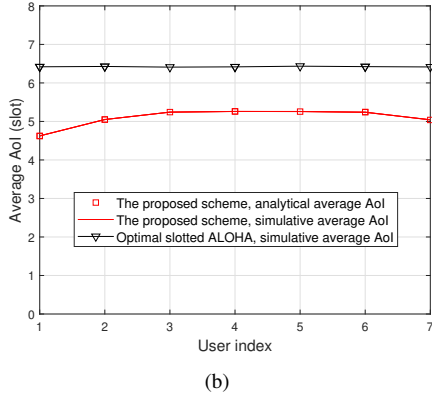
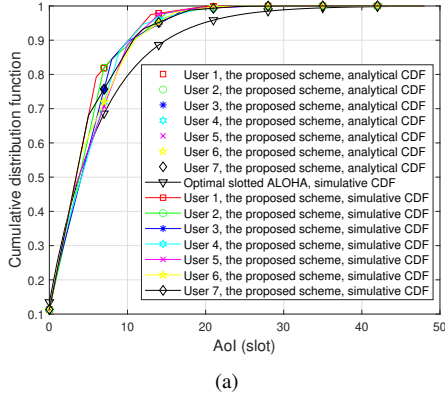


Fig. 1. CDFs of AoI and average AoI of the proposed scheme and optimal slotted ALOHA under $\gamma = 2$ and $M = 7$.

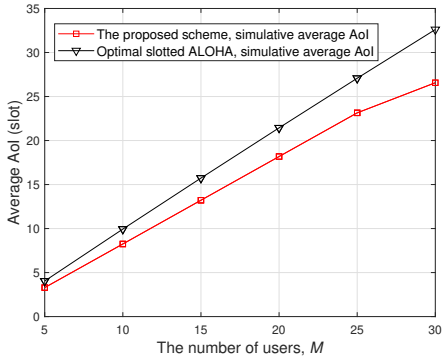


Fig. 2. Average AoI of the proposed scheme and optimal slotted ALOHA under $\gamma = 2$ and $M = 5, 10, \dots, 30$.

B. Comparison under $\gamma > 2$

In Fig. 3, we show the simulative average AoI under the proposed scheme and optimal slotted ALOHA for $\gamma = 3, 4, \dots, 7$ and $M = 17, 23$. As expected, we observe that both the schemes enjoy lower average AoI as γ increases owing to the enhanced MPR capability, but enjoy higher average AoI as M increases due to the severer contention level. We further observe that our proposed scheme outperforms optimal slotted ALOHA for all the cases, but the advantage decreases as γ

TABLE I
WORST-CASE AOI OF THE PROPOSED SCHEME AND OPTIMAL SLOTTED ALOHA UNDER $\gamma = 2$ AND $M = 5, 10, \dots, 30$.

The number of users, M	5	10	15	20	25	30
The proposed scheme	23	75	130	208	258	353
Optimal slotted ALOHA	94	191	310	413	499	657

increases. This advantage reduction can be attributed to the fact that a high MPR capability at the physical layer would alleviate the effect of scheme design at the MAC layer.

For $\gamma = 3, 4, \dots, 7$ and $M = 17, 23$, Table II shows that the proposed scheme produces much smaller worst-case AoI than optimal slotted ALOHA in all the cases. This verifies the effect of the UI property on the worst-case AoI again.

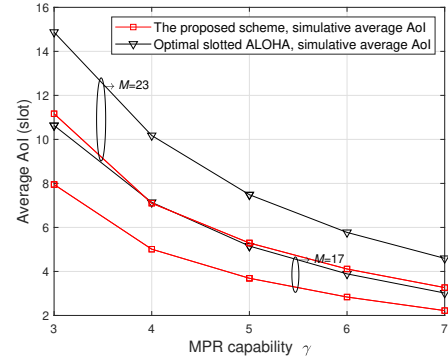


Fig. 3. Average AoI of the proposed scheme and optimal slotted ALOHA under $\gamma = 3, 4, \dots, 7$ and $M = 17, 23$.

TABLE II
WORST-CASE AOI OF THE PROPOSED SCHEME AND OPTIMAL SLOTTED ALOHA UNDER $\gamma = 3, 4, \dots, 7$ AND $M = 17, 23$.

M	Scheme	$\gamma = 3$	$\gamma = 4$	$\gamma = 5$	$\gamma = 6$	$\gamma = 7$
17	The proposed scheme	102	73	56	38	29
	Optimal slotted ALOHA	208	139	113	85	75
23	The proposed scheme	155	108	72	57	50
	Optimal slotted ALOHA	284	204	148	127	97

VI. CONCLUSIONS

In this paper, we have proposed protocol-sequence-based schemes under the MPR capability $\gamma \geq 2$. For $\gamma = 2$, using prime sequences [13] as protocol sequences, we have proved the UI property and derived the CDF of AoI and average AoI. For an arbitrary $\gamma > 2$, we have proposed a new construction based on CRT-UI sequences [14] to produce suitable protocol sequences under the considered γ , and proved its UI property. Simulation results showed that the proposed schemes outperform slotted ALOHA in terms of average AoI and worst-case AoI for various network settings.

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